Uptime Monitor — Database Schema

PostgreSQL 16 · 7 tables · captured from the live `uptime_monitor` database · 2026-07-08

Relationship map

```
users ──< urls ──< ping_history          (every check result)
  │         ├──< url_incidents          (downtime episodes)
  │         ├──< maintenance_windows    (planned downtime)
  │         └──< alert_deliveries >──┐  (audit log of alerts sent)
  └──────< alert_channels ────────────┘  (where to notify)
```

`──<` = one-to-many. Children cascade-delete with their parent, except `alert_deliveries.url_id` (SET NULL) so the alert audit log survives monitor deletion.

users

accounts & auth

Application accounts. Passwords stored as bcrypt hashes; role gates admin-only endpoints.

COLUMN	TYPE	NOTES
id	serial	PK
full_name	varchar(150)	NOT NULL
email	varchar(255)	NOT NULL · UNIQUE
hashed_password	varchar(255)	NOT NULL · bcrypt
role	varchar(10)	NOT NULL · 'viewer' | 'admin' · default 'viewer'
created_at	timestamp	default CURRENT_TIMESTAMP

urls

the monitors — core table

One row per monitored URL. Groups: identity, scheduling (Phase 1), and false-down state machine (Phase 2).

COLUMN	TYPE	NOTES
id	serial	PK
user_id	integer	FK → users(id) ON DELETE CASCADE
web_address	varchar(255)	NOT NULL · UNIQUE with user_id
name	varchar(100)	NOT NULL
status	varchar(20)	'PENDING' | 'UP' | 'WARN' | 'DOWN' · default 'PENDING'
check_type	varchar(160)	NOT NULL · e.g. HTTP, TTFB, SSL_EXPIRY, KEYWORD · default 'HTTP'
keyword_to_find	varchar(255)	for KEYWORD checks
check_interval_seconds	integer	NOT NULL · default 30
ping_interval_seconds	integer	NOT NULL · default 30
next_check_at	timestamptz	scheduling cursor — worker claims due rows (indexed)
last_pinged_at	timestamptz	last completed check
last_status_change_at	timestamptz	set on UP/DOWN transitions
consecutive_failures	integer	NOT NULL · default 0 · 2 ⇒ DOWN
consecutive_successes	integer	NOT NULL · default 0
last_error_reason	text	e.g. "HTTP 500", "Connection timed out"
is_public	boolean	shown on public status page · default false
created_at	timestamp	default CURRENT_TIMESTAMP
updated_at	timestamptz	NOT NULL · default now() · auto-stamped on every UPDATE by the `urls_set_updated_at` trigger

Indexes: `idx_urls_next_check_at` (drives the FOR UPDATE SKIP LOCKED claim query) · `urls_user_web_address_unique` ·
Trigger: `urls_set_updated_at` (BEFORE UPDATE → `set_updated_at()`)

ping_history

raw check results

One row per executed check. Source for uptime %, latency charts, and the metrics API.

COLUMN	TYPE	NOTES
id	serial	PK
url_id	integer	FK → urls(id) ON DELETE CASCADE
checked_at	timestamp	UTC · default CURRENT_TIMESTAMP
response_time_ms	integer	NULL on failure
status_code	integer	HTTP status, when applicable
is_up	boolean	NOT NULL
check_type	varchar(20)	which probe produced this row
extra_data	jsonb	per-check detail (status code, error, cert days…)

url_incidents

downtime episodes

Opened when a monitor confirms DOWN (2 consecutive failures), resolved on recovery. Powers incident timeline and downtime-minutes metrics.

COLUMN	TYPE	NOTES
id	serial	PK
url_id	integer	FK → urls(id) ON DELETE CASCADE · NOT NULL
started_at	timestamptz	NOT NULL
resolved_at	timestamptz	NULL = still open
check_type	varchar(20)	NOT NULL · which check failed
severity	varchar(10)	NOT NULL
acknowledged_at	timestamptz	operator ack
note	text	postmortem note

Index: `idx_url_incidents_open` — partial index `WHERE resolved_at IS NULL` keeps the active-incidents query fast.

alert_channels

notification targets

Per-user notification endpoints. A user's enabled channels fire for all of that user's monitors.

COLUMN	TYPE	NOTES
id	serial	PK
user_id	integer	FK → users(id) ON DELETE CASCADE · NOT NULL
channel_type	varchar(20)	NOT NULL · 'EMAIL' | 'WEBHOOK'
name	varchar(100)	NOT NULL · display label
destination	text	NOT NULL · email address or webhook URL
headers	jsonb	NOT NULL · default '{}' · custom key/value HTTP headers (Authorization, X-API-Key…)
http_method	varchar(10)	NOT NULL · 'POST' | 'GET' | 'PUT' | 'PATCH' · default 'POST'
notify_on_down	boolean	NOT NULL · default true
notify_on_recovery	boolean	NOT NULL · default true
is_enabled	boolean	NOT NULL · default true
created_at	timestamptz	default now()

alert_deliveries

alert audit log

One row per send attempt (success or failure) — makes alerting provable in the UI.

COLUMN	TYPE	NOTES
id	serial	PK
channel_id	integer	FK → alert_channels(id) ON DELETE CASCADE
url_id	integer	FK → urls(id) ON DELETE SET NULL (log survives deletion)
incident_id	integer	related incident, if any
event_type	varchar(10)	NOT NULL · 'DOWN' | 'UP' | 'TEST'
status	varchar(10)	NOT NULL · 'SENT' | 'FAILED'
error	text	failure reason, truncated to 500 chars
created_at	timestamptz	default now()

Index: `idx_alert_deliveries_channel` on `(channel_id, created_at DESC)` for the recent-deliveries panel.

maintenance_windows

planned downtime

Scheduled maintenance shown on the public status page.

COLUMN	TYPE	NOTES
id	serial	PK
url_id	integer	FK → urls(id) ON DELETE CASCADE
title	varchar(255)	NOT NULL
message	text	subscriber-visible description
starts_at	timestamptz	NOT NULL
ends_at	timestamptz	NOT NULL

Known design wart: older tables (`users.created_at`, `urls.created_at`, `ping_history.checked_at`) use naive `TIMESTAMP` (UTC by convention) while all newer columns use `TIMESTAMPTZ`. This split caused two timezone display bugs (both fixed in the frontend's `parseApiDate` and the metrics API). A future migration could normalize everything to `TIMESTAMPTZ`.